Unit 6 Practice — Functions & Linear Relationships — Answer Key

Functions · Slope · Tables → Equations · Real-World · Comparing · Qualitative Graphs · Systems

A. Functions

Problem 1 Score 3

Does the table shown represent a function? (yes/no)

x	y
1	5
2	7
3	9
4	11

Answer: yes

Problem 2 Score 3

Does the table shown represent a function? (yes/no)

x	y
2	4
3	6
2	9
5	8

Answer: no

Problem 3 Score 4

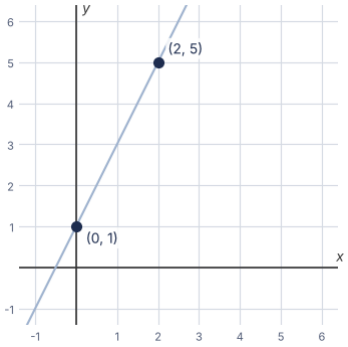
A relation includes the points (1,3), (2,5), (3,7), (1,9). Is it a function?

Answer: no

B. Slope

Problem 1 Score 3

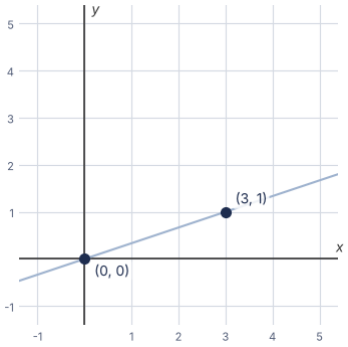
Find the slope of the line through the two points shown.



Answer: 2

Problem 2 Score 4

Find the slope of the line through the two points shown.



Answer: $\frac{1}{3}$

C. Tables → Equations

Problem 1 Score 4

Write the equation (y = mx + b) for the table shown.

x	y
0	1
1	3
2	5
3	7

Answer: y=2x+1

Problem 2 Score 5

Write the equation (y = mx + b) for the table shown.

x	y
0	10
1	7
2	4
3	1

Answer: y=-3x+10

Problem 3 Score 5

A line passes through (2, 7) and (6, 15). Use algebra to find its y-intercept.

Answer: 3

D. Real-World Applications

Problem 1 Score 4

A taxi charges a $4 flat fee plus $2 per mile. Write the cost equation (y = mx + b) where x = miles.

Answer: y=2x+4

Problem 2 Score 4

A gym charges $30 to join plus $25/month. How much for 6 months total?

Answer: 180

Problem 3 Score 5

A candle is 20 cm tall. After 4 hours it is 14 cm tall. It burns at a steady rate. Which equation gives its height y (in cm) after x hours?

A. y = 1.5x + 20
B. y = −6x + 20
C. y = −1.5x + 14
D. y = −1.5x + 20

Answer: D

E. Comparing Functions

Problem 1 Score 4

Function A: y = 3x + 2. Function B has a table showing slope 5. Which has the greater rate of change?

Answer: function b

Problem 2 Score 5

Function A: y = 4x − 1. Function B passes through (0,3) and (2,7). Which has the greater y-intercept?

Answer: function b

Problem 3 Score 5

Plan A costs $30 to start plus $10 each month. Plan B costs nothing to start, and its line passes through (0, 0) and (2, 50). After how many months do the two plans cost the same?

Answer: 2

F. Qualitative Graphs

Problem 1 Score 3

A graph shows temperature rising steadily, then staying flat for an hour, then rising again. During the flat part, is temperature increasing, decreasing, or constant?

Answer: constant

Problem 2 Score 3

A car's distance-time graph shows a steep upward line, then a horizontal line. What is happening during the horizontal part?

A. Stopped
B. Speeding up
C. Slowing down
D. Going back home

Answer: stopped

G. Systems of Equations

Problem 1 Score 3

Find the solution (x,y) to the system of equations graphed.

Answer: (2,4)

Problem 2 Score 5

Solve the system: y = x + 2 and y = 3x − 2. Find x.

Answer: 2